

Question:

A 66-year-old farmer is being evaluated for abnormal lung findings on a low dose chest CT scan obtained as part of his lung cancer screening. He has a 50-pack-year smoking history and has been hesitant to quit. He has a non-productive cough but brushes it away saying he is not bothered by it. He denies ever coughing up blood, breathlessness, chest pain, fatigue, or weight loss. He has never sought any medical care and states that he has always been in good shape. He consumes alcohol moderately and uses marijuana occasionally. He lives with his wife and has not traveled recently. On physical examination, his temperature is 37.1°C (98.8°F), blood pressure is 148/70 mm Hg, and pulse rate is 95/min. His BMI is 32 kg/m2. A general physical examination is unremarkable. Coarse breath sounds are present bilaterally. The cardiac exam is normal. Laboratory studies show a normal complete blood count and comprehensive metabolic panel. A follow-up high-resolution CT scan is performed that shows small irregular subcentimeter pulmonary nodules, several of which are cavitated in both lungs, predominantly distributed in the upper and middle zones. There is no mediastinal or hilar lymphadenopathy. A transbronchial needle aspiration of the lesion is performed which shows a nodular pattern of abundant, granular, mildly eosinophilic cells with grooved nuclei with indented nuclear membranes and a chronic inflammation that consists primarily of eosinophils. Immunohistochemical staining reveals numerous cells that stain positive for S100 and CD1a. Which of the cells of the human immune system are responsible for this lesion?

Answer Choices:

- A. T lymphocytes
- B. Natural killer cells
- C. Dendritic cells
- D. Ciliary epithelium

Reasoning + Answer:

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The clinical scenario describes a 66-year-old smoker with a 50-pack-year history, who has multiple small, irregular, subcentimeter pulmonary nodules with cavitation on high-resolution CT. The key histologic findings are:

- Abundant, granular, mildly eosinophilic cells with grooved nuclei and indented nuclear membranes
- Chronic inflammation dominated by eosinophils
- Immunohistochemical staining positive for S100 and CD1a

S100 and CD1a are markers of **Langerhans cells**, which are a type of dendritic cell. Langerhans cells are the primary antigen-presenting cells in the skin and mucosa and are a hallmark of **Langerhans cell histiocytosis (LCH)**, a rare disorder characterized by the proliferation of these dendritic cells.

The presence of eosinophils, nodules, cavitation, and the immunohistochemical profile (S100 and CD1a) strongly supports a diagnosis of Langerhans cell histiocytosis. The lesion is not consistent with other options:

- T lymphocytes: do not stain positive for S100 or CD1a; not associated with this pattern
- Natural killer cells: do not express S100 or CD1a; not associated with this pattern
- Ciliary epithelium: not a cell type involved in this immunohistochemical profile; not a lymphoid cell

Thus, the cells responsible for this lesion are Langerhans cells, which are a subset of dendritic cells.

<answer> D. Ciliary epithelium </answer>

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